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Collision risk analysis of mega constellations in low Earth orbit

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Abstract

The LEO megaconstellations have thousands of satellites, which operate on similar orbital heights. Because of increasing space debris, the satellites accelerate the growth of the number of space objects, increase the threat to their own safety and sustainable utilization of the space environment. This paper focuses on short-term collision probability and long-term collision probability calculation model for collision risk of mega constellations. On the one hand, aiming at how to solve the problem quickly and accurately for rendezvous collisions, a short-term collision risk evolution calculation model of LEO mega-constellation is established. By investigating the space collision target screening methods of large-scale satellites against large-scale space debris, including apogee–perigee filtering, time window filtering, space discrete volume element filtering and relative position filtering, the filtered target trajectory is fitted by Lagrange interpolation method to obtain the motion state at the closest time. Then, the collision probability calculation method of Laplace transform is used to calculate the collision probability of the target at the closest time. On the other hand, the long-term collision risk evolution model of LEO Mega constellation is established. By simplifying the model of debris and collision, we explore the calculation method of space object density, space debris attenuation model, space target disintegration model and the distribution of disintegration targets are studied. According to the way of space debris' generation and extinction to calculate the change of space density. The collision probability between the satellite and space debris and space debris are calculated by using the calculation method of collision probability of particle-in-a-box (PIB). Finally, through the simulation of Starlink phase I constellation, results show that after the deployment of the Starlink constellation, the probability of a short-term collision in the constellation shell increases by 30%–40%, the probability of at least one collision in the constellation lifetime is 70.2%, and the probability of a secondary collision increases by 25.3% after a collision.

This study provides a theoretical reference for analyzing the orbital safety of the LEO mega-constellation, and provides a reference for the safe operation of spacecraft on-orbit and the sustainable development and utilization of space resources.

Keywords: Low Earth Orbit; mega constellation; space debris; collision probability; collision risk

1. Introduction

The increase in human space activities is the main factor leading to the gradual severity of the spacecraft safety situation. With the increasing number of satellites in space, collisions, explosions and failures may occur during long-term operation, which are often accompanied by a large number of space debris, resulting in a further increase in the density of space objects and further aggravating the risk of satellite collisions. If such a vicious cycle is not controlled, it may eventually produce the Kessler syndrome [1], causing the spacecraft to lose its orbital environment.

The reference[2] shows that as of June 6, 2023, there has been 36,500 space debris objects greater than 10 cm in space and 1,000,000 space debris objects from greater than 1 cm to 10 cm. Number of debris objects regularly tracked by Space Surveillance Networks and maintained in their catalogue about 33470. These cataloged targets are distributed in orbits at various altitudes, and most of them pose no security threat to the orbit in which the low-orbit megaconstellation is located. Therefore, in order to reduce the amount of calculation and facilitate accurate and fast collision probability calculation, it is necessary to screen out targets that do not pose a security threat to the LEO megaconstellation as much as possible.

In recent years, the concept of commercial Internet satellite platform has been proposed, resulting in a wave of mega constellation construction. As of May 2023, Starlink has launched 4521 satellites and Oneweb has launched 636 satellites. Based on data from the International Telecommunications Union (ITU), Federal Communications Commission (FCC), and Jonathan's Space Pages, many companies have proposed construction plans of mega constellation in low Earth orbit (LEO)[3]. These mega constellations contain a large number of small satellites, and they are all in the same region, with a given width. If the deployments are completed in accordance with the plan, then the number of satellites in orbit will be several times the sum of the number of past launches. Table 1 gives some mega constellation plans in LEO.

Table 1 Some mega constellation plans in LEO[3-6]

Constellation	Orbit altitude/km	Satellite number
OneWeb	1200	7020
Starlink	328–614	41926
Kuiper	590–630	3236
GW	508–1145	12992
Telesat	1015–1325	1671
Boeing/LEO	1200	2956

Therefore, whether from the perspective of the safety of the LEO megaconstellation itself or the sustainability of the space environment, the research on the orbital safety of the low-orbit megaconsion, especially the collision safety, is necessary and urgent.

The mega constellations in LEO represented by the Starlink constellation have begun to be deployed. From 2019 to April 2023, Starlink showed frequent close proximity during the first phase of deployment. Starlink-44 had a close approach with Euro-Aeolus satellites in 2019[7]. About 1600 close proximity events occurred in 2021, accounting for half of all space proximity events of the year[8, 9].

With the deployment of the constellation program, the environment in LEO space is becoming increasingly severe. Taking the OneWeb constellation as an example, Jonas et al. studied the interaction between the space debris environment and the mega constellations[10] , where the influence of LEO mega constellations on the space environment was investigated[11, 12]. Lorenzo designed a simple risk assessment statistical tool for mega constellations in the LEO space debris environment to analyze the influence of the main parameters of the constellation on their vulnerability[13]. In addition, Isoletta proposes a uncertainty aware Cube mid-term collision risk assessment algorithm, which is capable of evaluating the target mid-term collision frequency in low orbit[14]. Adilov et al. estimated the expected annual cost of a satellite's collision with orbital debris based on orbital debris economics[15].

Due to the limited accuracy of the tracking data used to assess the risk of collisions, the maneuvers currently being performed far exceed the necessary level, which leads to overestimates. Campiti proposes a solution for equipping satellites with sensors that will enable them to autonomously track

dangerous objects when at risk of collision and refine their location knowledge before making maneuvering decisions, enabling data collection and estimating collision risk, thereby improving the accuracy of final predictions and reducing false alarms[16].

Lewis [17] introduced a large constellation to analyze the influence of low-orbit mega constellation deployment on the collision probability of space debris. Under the condition that the post mission disposal criteria are met, the amount of space debris will continue to grow. Le May used the MASTER-2009 debris evolution model to simulate the OneWeb and SpaceX constellations, and further analyzed the influence of constellation parameters and the long-term influence of satellites that failed to perform deorbit operations[18]. Rossi utilized different tools developed by IFAC-CNR to analyze the short-term to long-term constellation influences for determining the effects associated with constellation deployment and execution[19]. For the long-term sustainable use of the space environment, many scholars have proposed debris mitigation plans for mega constellations[20, 21].

Eduardo proposed an analysis model for evaluating the collision probability between a space object and a satellite constellation when it is out of orbit or injected into orbit. The consequences of in orbit collisions are also investigated by assessing the collision risk caused by the formation of debris clouds[22]. Mega constellation satellites may disintegrate and generate hundreds or thousands of space debris if one collision occurs due to their small mass and fast running speed. The debris generated by the collision can stay in space for a long time and may cause secondary damage to the constellation[23].

However, spacecraft and space debris are mainly concentrated in the low-Earth orbit region, and if the traditional two-pair analysis method is used to solve the intersection collision of the two targets, the calculation will be very large. This paper focuses on how to solve the problem quickly and accurately for rendezvous collisions, and uses a sequence of filters to greatly reduce the computational expense of the algorithm. On this basis, to evaluate the collision risk of mega constellations（especially the Starlink constellation）, a short-term collision probability calculation model and a long-term collision probability calculation model are established. These models are used to calculate the collision risk of the Starlink constellation.

The main contributions of this work are summarized as follows:(1) The proximity analysis method of large space debris and large satellite was proposed, and the space collision target filtering method for large-scale space debris by large-scale satellites was studied. (2) A collision evolution calculation method between large-scale space debris and large-scale satellites is proposed. By filtering and partitioning data, the computing efficiency is improved while satisfying high accuracy. (3) Aiming at the long-term collision risk problem after the deployment of low-orbit megaconstellations, a long-term collision risk calculation model of low-orbit megaconsions is established.

The remainder of the manuscript is organised as follows. In Section 2, the calculation methods for short-term and long-term collision probabilities are presented. Based on the proposed method, the short-term and long-term collision probabilities of the Starlink constellation are calculated, as presented in Section 3., Furthermoe, through comparison of the collision probability before constellation deployment, the increased collision risk percentage after constellation deployment is obtained. Finally, in Section 4, the conclusive remarks as well as future perspective are given.

2. Short-term collision probability calculation model

The orbital prediction model in this summary uses the SGP4 model to carry out short-term collision studies. This chapter establishes a proximity analysis method of large-scale space debris and large-scale satellites to eliminate the dangerous targets. The short-term collision probability computational model is constructed in accordance with the collision probability calculation method of the rendezvous target. This model is used to calculate the short-term in-orbit natural evolution collision in the shell where the

constellation is located. A series of methods are used for proximity analysis of space targets, as shown in Figure 1.

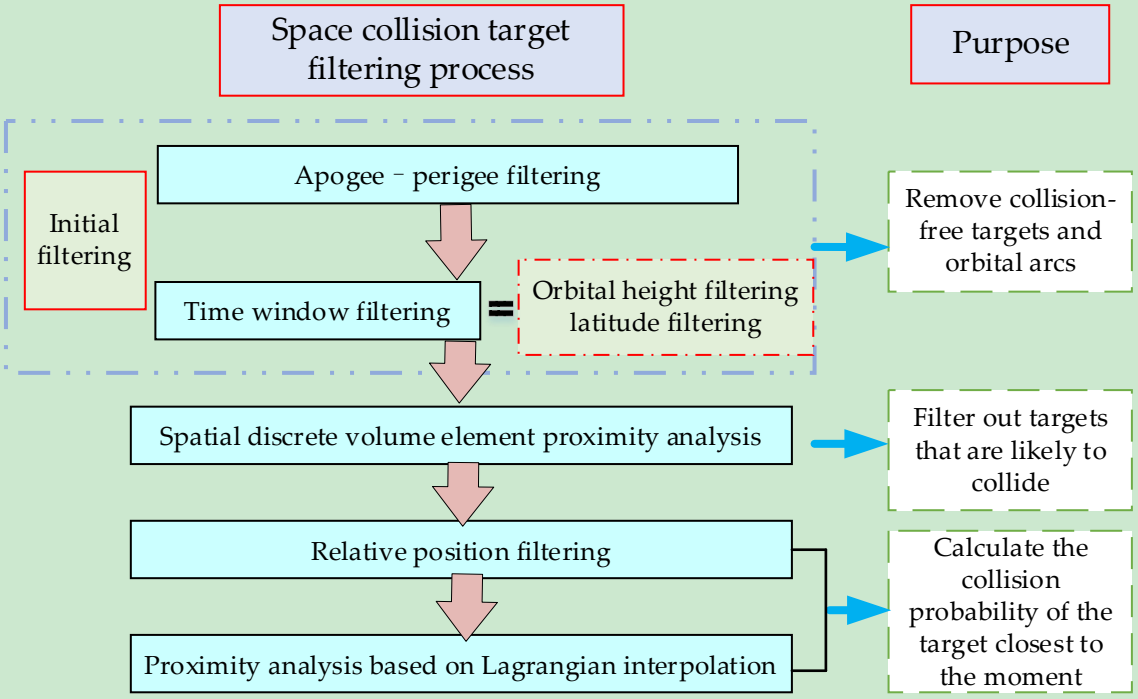


Figure 1. Space targets proximity analysis process

2.1 Apogee–perigee filtering

When the orbital altitude of the LEO mega constellation is determined, the constellation satellites will operate in the constellation shell. At this time, the orbit of the space object and the shell have three states: intersection, inclusion, and separation. Space objects with orbits intersecting the shell or are contained within the shell may collide with the constellation satellites, and space objects with orbits separated from the shell will not collide with the constellation. Therefore, the targets that may collide with the constellation shell can be preliminarily screened in accordance with the geometric relationship for the apogee and perigee of the target orbit. The relationship between the apogee–perigee and the shell of the space target is shown in Figure 2.

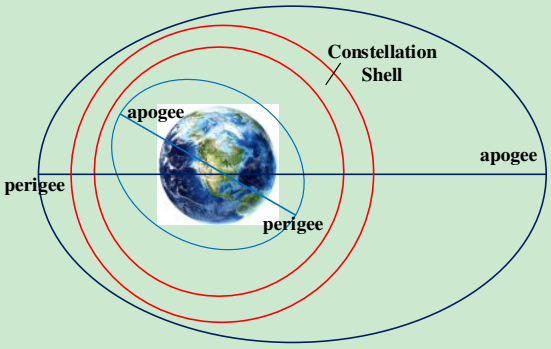


Figure 2. Schematic of the relationship between the space target and the shell.

Assuming that the height of the constellation shell is D, the nominal orbital semi major axis of the constellation satellites is a , and the apogee and perigee of the space target ^{s} are $Apog_s$ and $Perg_s$, respectively. Given that the eccentricity of the LEO mega constellation satellites is approximately 0, the nominal orbital radius of the constellation can be approximately equal to a . As shown in Figure 2, the apogee of the space target is compared with the upper of the constellation shell, and the perigee of the space target is compared with the lower bounds of the constellation shell. If the perigee is greater than the radius of the upper bound of the constellation shell or the apogee is smaller than the radius of the lower bound of the constellation shell, then the target is not a threat and should be eliminated. The apogee–perigee filtering method is expressed as

$$Perg_s > a + D/2 \| Apog_s < a - D/2 \tag{1}$$

2.2 Time window filtering

The space targets may collide with the constellation satellite only when they travel into the shell of the constellation. Thus, the orbital altitude of the space target position determines whether it may collide with a constellation satellite. When the height of the space target position is outside the range of the shell height, the target will not collide with the constellation satellite. The constellation satellite may collide with the target only when the space target is within the shell range.

Assuming that the position and velocity of the space target at moment t is $\rho_s(\vec{r}(t),\vec{v}(t))$, the radius of the space target at the moment is:

$$r(t) = \sqrt{\vec{r}_x(t) + \vec{r}_y(t) + \vec{r}_z(t)} \tag{2}$$

Given that the height interval of the constellation shell is $[a - D/2, a + D/2]$, the time window for the possible collision between the space target and the constellation satellite can be expressed as

$$a - D/2 < r(t) < a + D/2 \tag{3}$$

Different LEO mega constellations have different orbital inclinations, and the latitude ranges that the constellations can cover are also different. Space targets may be higher than the latitudes that the constellation can cover when passing through the shell. Therefore, whether the latitude is within the latitude range covered by the constellation when the satellite is in the constellation shell should be continuously analyzed.

In accordance with the integral formula of orbital area, the moment vector of momentum per unit mass $\vec{h}$ is expressed as

$$\vec{h} = \vec{r}(t) \times \vec{v}(t) = h_x i + h_y j + h_z k \tag{4}$$

where h_x, h_y, h_z are the projected sizes of the moment of momentum per unit mass in the directions of the three coordinate axes. Given that the orbital inclination i_s of the space target s is the angle between the momentum moment vector and the Earth's north pole, we obtain

$$\cos i_s = h_z / h \tag{5}$$

where h is the modulus of the unit vector moment of momentum. We can obtain the positional relationship by projecting the satellite orbit on the ground, as shown in Figure 3.

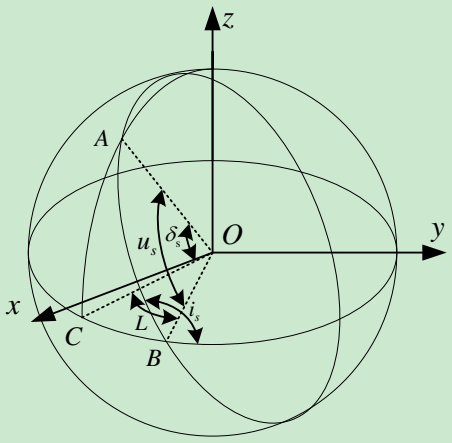


Figure3. Projection of the space target orbit on the ground.

In accordance with the spherical triangle, we can obtain

$$\delta_s = \arcsin(\sin i_s \sin u_s) \tag{6}$$

where u_s is the phase of the space target s, and δ_s is the latitude of the space target s. When the latitude of the space target s within the window range is greater than the sum of the orbital inclination i of the constellation and the offset of the orbital inclination Δi, the space target will not collide with the constellation satellite and should be eliminated. The filtering is expressed as

$$\delta_s > i + \Delta i \tag{7}$$

2.3 Spatial discrete volume element proximity analysis

The probability of collision between space targets and constellation satellites after apogee–perigee filtering and time window filtering is high. If the pairwise calculation method is used to determine whether two targets collide, then the computational complexity will vary in a second-order growth with the number of constellation satellites and the number of space targets. Thus, the collision probability calculation of LEO mega constellations is unsuitable. In this chapter, the constellation shell is divided into several space volume elements, and the objects that may collide are filtered by judging the distance among the space objects in each volume element. At this time, the computational complexity is in linear growth with the number of constellation satellites and the number of space objects. This condition can greatly reduce the computational complexity of collisions between LEO mega constellations and space debris, and improve computational efficiency.

The space target orbit used in this study is defined under the J2000 inertial coordinate system, so the discrete volume element coordinate system is also defined as the J2000 inertial coordinate system. In the inertial coordinate system, the constellation shell is centered on the center of the Earth, and the spherical coordinates are established along the three directions of geocentric distance, longitude, and latitude. The discrete volume element is the starting point $r_{\min}$、$\lambda_{\min}$、$\varphi_{\min}$, and the space of the constellation shell is divided into several space volume elements in accordance with the interval $\Delta h, \Delta\lambda, \Delta\varphi$. Figure 4 shows a schematic of a space volume element.

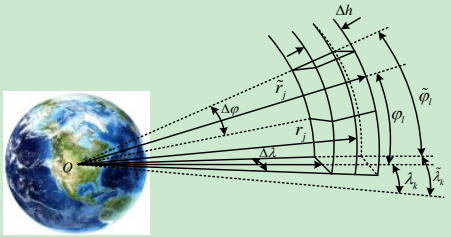


Figure 4. Schematic of space discrete volume element.

In accordance with Figure 4, the coordinates of the space volume element in three directions are expressed as

$$\begin{aligned} \tilde{r}_j &= r_{\min} + j\Delta h,\ j = 0,1,2...,J \\ \tilde{\lambda}_k &= \lambda_{\min} + k\Delta\lambda,\ j = 0,1,2...,K \\ \tilde{\varphi}_l &= \varphi_{\min} + j\Delta\varphi,\ j = 0,1,2...,L \end{aligned} \tag{8}$$

The center of the space volume element is expressed as

$$\begin{aligned} r_j &= \tilde{r}_j - \Delta h/2,\ j = 1,2...,J \\ \lambda_k &= \tilde{\lambda}_k - \Delta\lambda/2,\ j = 1,2...,K \\ \varphi_l &= \tilde{\varphi}_l - \Delta\varphi/2,\ j = 1,2...,L \end{aligned} \tag{9}$$

The volume element division needs to consider factors, such as calculation amount, storage amount, and accuracy. The larger the volume element, the less the number of traversed volume elements, the clearer the trajectory of the space target in the volume element, and the greater the number of targets appearing in the same volume element at the same time, resulting in an increase in the computational complexity of collision discrimination between targets. The smaller the discrete volume element, the fewer objects appear in the same volume element, the more time it takes to traverse the volume element, and the trajectory points of spatial objects in the volume element may be skipped, resulting in a decrease in computational accuracy. Therefore, setting the volume element size reasonably becomes the key to the approach analysis.

The orbital planes of LEO mega constellations are uniformly distributed along the right ascension of ascending node direction, and the phase and latitude differences between satellites in the orbital plane are equal. A parameter slightly smaller than the reference standard is selected to divide the volume element while ensuring a small number of LEO mega constellation satellites in the volume element. The volume of the volume element is increased, so that the trajectory continuity of the space target in the volume element is stronger. Given that the size of volume elements decreases along the latitudinal direction, the common method is to introduce the maximum latitude and combine the volume elements with the latitude greater than ℅ that in each altitude layer into the same volume element to deal with the volume elements near the poles of the constellation shell.

2.4 Relative position filtering

When the space target is in the same volume element, it is judged whether a collision occurs between satellites by the approach distance. Given that the attitude of the space target is unknown, the space target is calculated as an envelope sphere. When the distance between the centers of mass of the two targets is less than the joint radius R, the two targets are considered to have collided immediately.

The true position of the space target has a random deviation, so the collision between the two space targets is a probabilistic event. In accordance with the "3σ" rule, the true position of the space target has a high probability of being within 3 standard deviations of the average position. Therefore, we can eliminate the targets that may collide for the next calculation in accordance with the relative position of the two space targets and the "3σ" rule.

Assuming that the position error square matrix of the two space targets in the orbital coordinate system is

$$\mathrm{var}(r_1)=\begin{bmatrix}\sigma_{N_1}^2 & & \\ & \sigma_{T_1}^2 & \\ & & \sigma_{W_1}^2\end{bmatrix},\ \mathrm{var}(r_2)=\begin{bmatrix}\sigma_{N_2}^2 & & \\ & \sigma_{T_2}^2 & \\ & & \sigma_{W_2}^2\end{bmatrix} \tag{10}$$

The variance matrix of the relative position vectors of the two targets in the Earth inertial coordinate system is

$$\mathrm{var}(r_{ECI})=\mathrm{var}(r_1-r_2)=M_{NTW\to ECI}(\mathrm{var}(r_1)+\mathrm{var}(r_2))M_{NTW\to ECI}^{T} \tag{11}$$

In accordance with the "3σ" rule, when two space targets may collide, the relative distance between the two targets satisfies the following conditions.

$$\left|\vec{r}_2-\vec{r}_1\right|<3\left|\sigma_{r_{ECI}}\right|+R \tag{12}$$

where r_2 and r_1 are the position vectors of the space target in the Earth inertial coordinate system. $\sigma_{r_{ECI}}$ is the joint error distance in the Earth inertial coordinate system.

2.5 Proximity analysis based on Lagrangian interpolation

Two space targets filtered out by relative positions may collide. Given that the two targets are most likely to collide when the orbits are closest, the collision probability at the closest orbit is regarded as the probability of collision between the two space targets. In reality, orbit prediction usually requires a certain step size. The step size cannot be infinitely small, and the trajectory cannot be continuous. Therefore, the predicted trajectory point is not necessarily the closest position of the orbit. Interpolation in accordance with the trajectory of the space target in the space cube must be performed to fit the curve equation of the orbit and find the position closest to the moment.

Orbits are usually fitted by using Lagrangian interpolation. Suppose the moment when the two targets in the space volume element are closest to the trajectory point is t_i, let the position vector of one of the space targets be $\vec{r}(t_i)$. Two trajectory points are selected before and after the closest trajectory point, and five points with the closest trajectory point are used as interpolation points to perform polynomial interpolation. The constructor is obtained in accordance with the Lagrangian interpolation method.

$$f_k(t)=\prod_{j\neq k}^{1\le j\le 5}\frac{(t-t_j)}{(t_k-t_j)} \tag{13}$$

The Lagrangian interpolation fitting curve equation of the target orbit can be obtained as:

$$\vec{f}(t)=\sum_{k=1}^{5}\vec{r}(t_k)f_k(t) \tag{14}$$

Given that the fitting curve is a quartic polynomial and the interpolation points are known, the Lagrangian interpolation fitting curve equation can be expressed as:

$$\vec{f}(t)=\vec{a}_0+\vec{a}_1t+\vec{a}_2t^2+\vec{a}_3t^3+\vec{a}_4t^4 \tag{15}$$

In accordance with Equation 15, the equation system of the linear curve is obtained as

$$\begin{pmatrix} \vec{r}_1 \\ \vec{r}_2 \\ \vec{r}_3 \\ \vec{r}_4 \\ \vec{r}_5 \end{pmatrix} = \begin{pmatrix} 1 & t_1 & t_1^2 & t_1^3 & t_1^4 \\ 1 & t_2 & t_2^2 & t_2^3 & t_2^4 \\ 1 & t_3 & t_3^2 & t_3^3 & t_3^4 \\ 1 & t_4 & t_4^2 & t_4^3 & t_4^4 \\ 1 & t_5 & t_5^2 & t_5^3 & t_5^4 \end{pmatrix} \begin{pmatrix} \vec{a}_0 \\ \vec{a}_1 \\ \vec{a}_2 \\ \vec{a}_3 \\ \vec{a}_4 \end{pmatrix} \tag{16}$$

In accordance with the above formula, the expression $\vec{f}(t)$ of the space target trajectory fitting equation can be obtained. The same method can be used to obtain the space target velocity fitting equation expression $\dot{\vec{f}}(t)$. The relative distance vector $\vec{r}_{rel}(t_i)$ of the two targets is:

$$\vec{r}_{rel}(t_i) = \vec{r}_2(t_i) - \vec{r}_1(t_i) \tag{17}$$

The relative position vector $\vec{f}_{rel}(t)$ over time can be expressed as

$$\vec{f}_{rel}(t) = \vec{f}_2(t) - \vec{f}_1(t) \tag{18}$$

The relative distance is the closest moment τ_C when it reaches the minimum value, which can be expressed as

$$\frac{d\left|\vec{f}_{rel}(t)\right|}{dt}\Big|_{t=\tau_C} = 0 \tag{19}$$

In accordance with the above formula, the closest time in the range of five trajectory points can be obtained, and the motion state of the two collision targets in space at the intersection point can be obtained by bringing $t = \tau_C$ into $\vec{f}_1(t)$, $\vec{f}_2(t)$, $\dot{\vec{f}}_1(t)$, and $\dot{\vec{f}}_2(t)$.

2.6 Collision probability calculation model

The position error of the encounter target is transformed into the encounter coordinate system, and the probability density function of the *3D* Gaussian distribution forming the joint error ellipsoid is

$$\rho(\mathbf{X}_e) = \frac{1}{\sqrt{(2\pi)^3 \det C_c}} e^{-\frac{1}{2}(\mathbf{X}_e - \mu_c)^T C_e^{-1} (\mathbf{X}_e - \mu_c)} \tag{20}$$

The joint position error ellipsoid is projected to the encounter plane, and the encounter coordinate system is rotated as the calculation coordinate system, so that the x_e axis of the calculation coordinate system is consistent with the direction of the major semi axis of the joint position error ellipse. The collision probability by integrating the joint position error ellipse domain in the calculation coordinate system is expressed as:

$$P_c = \iint\limits_{x^2+y^2 \le R_c^2} \frac{1}{2\pi\sigma_x\sigma_y} \exp[-\frac{1}{2}(\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2}))dxdy \tag{21}$$

where R_c is the effective radius of the joint circle domain of the two targets, and μ_x、μ_y、σ_x、σ_y are the mean and variance of the position error of the joint error ellipse domain in x axis and y axis directions of the calculated coordinate system.

Equation 21 transforms the collision probability calculation into a 2D probability density function integration problem in a circular domain. The commonly used methods for this integral are numerical and analytical methods. The numerical method has a large amount of calculation and is unsuitable for large-scale targets, whereas the analytical method has fast calculation speed and high calculation accuracy, which can meet the calculation requirements. However, the traditional Chan method approximates the integral domain, and the obtained new integral model has a deviation that cannot be accurately measured from the original initial integral model[20]. Reference [21] improves the defects in the Chan method and proposes a Laplace transform-based method. The space target collision probability calculation method improves the calculation accuracy and speed. Therefore, this method is used to solve the collision probability in this study.

$$P_c = e^{(-pR^2)} \cdot \sum_{k=0}^{+\infty} \frac{a_k R^{2(k+1)}}{(k+1)!} \tag{22}$$

$$a_0 = \frac{1}{2\sigma_x\sigma_y}\exp(-\frac{1}{2}(\frac{\mu_x^2}{\sigma_x^4}+\frac{\mu_y^2}{\sigma_y^4})) \tag{23}$$

$$a_1 = a_0 p\left(1+\frac{\eta}{2}+\frac{Q_x+Q_y}{p}\right) \tag{24}$$

$$a_2 = \frac{a_0}{2}p^2[(1+\frac{\eta}{2}+\frac{Q_x+Q_y}{p})^2+(1+\frac{\eta^2}{2}+\frac{\eta Q_x}{p})] \tag{25}$$

$$\begin{aligned} a_3 = {} & \frac{a_0}{6}p^3[(1+\frac{\eta}{2}+\frac{Q_x+Q_y}{p})^3+3(1+\frac{\eta}{2}+\frac{Q_x+Q_y}{p})+(1+\frac{\eta^2}{2}+\frac{2\eta Q_x}{p})]+ \\ & \frac{a_0}{6}p^3[2(1+\frac{\eta^3}{2}+\frac{3\eta^2 Q_x}{p})] \end{aligned} \tag{26}$$

$$\begin{aligned} a_{k+4} = {} & \frac{1}{k+4}\left(-p^3\eta^2 Q_y a_k + p^2\eta\left(p\eta\left(k+\frac{5}{2}\right)+2Q_y\left(\frac{\eta}{2}+1\right)\right)a_{k+1}\right)- \\ & \frac{p}{k+4}\left(p\eta\left(\frac{\eta}{2}+1\right)(2k+5)+\eta\left(2Q_y+\frac{3p}{2}\right)+Q_x+Q_y\right)a_{k+2}+ \\ & \frac{1}{k+4}\left(p(2\eta+1)(k+3)+p\left(\frac{\eta}{2}+1\right)+Q_x+Q_y\right)a_{k+3} \end{aligned} \tag{27}$$

where R is the radius of the two target joint spheres, p is the undetermined factor, and the other parameters are:

$$\eta = 1-\frac{\sigma_y^2}{\sigma_x^2} \quad Q_x = \frac{\mu_x^2}{4\sigma_x^4} \quad Q_y = \frac{\mu_y^2}{4\sigma_y^4} \tag{28}$$

where μ_x and μ_y are the expected error values; σ_x and σ_y are the standard deviations of the error in the calculated coordinate system.

3 Long-term collision probability calculation model

The long-term orbit safety of LEO mega constellations is not only complex and time-consuming in calculation, but also requires high modeling and hardware conditions. Moreover, due to the influence of orbit prediction errors, there is a significant deviation between the predicted position of the long-term orbit of space targets and the actual position, and the calculation results still have a large random deviation.

This section makes simplified assumptions about the debris and collision models, and calculates the density change of space debris in the shell where the constellation is located in accordance with the generation and death of space debris. This process is performed to avoid detailed modeling and analysis of complex scenarios for collisions in the life cycle of the constellation. A “particle-in-a-box” (PIB) collision probability calculation method is used to establish a long-term collision probability calculation model. The process of establishing the model is shown in Figure 5.

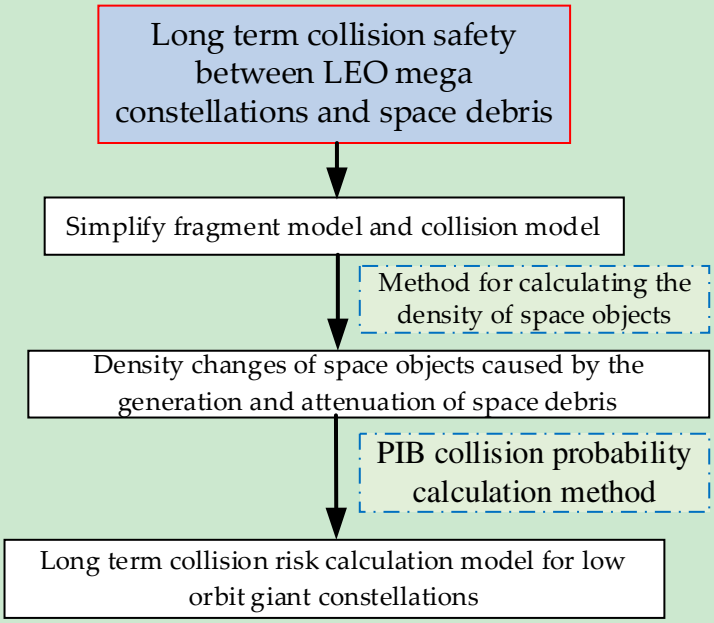


Figure 5. Long-term collision probability calculation model

3.1 Model simplifying assumptions

Educated guesses and rationalized assumptions about relevant variables and their ranges are required to simplify computational models[22]. The assumptions defined in this section are as follows:

(1) All spatial objects are treated as spheres.

(2) The objects in the initial environment of LEO are divided into two groups for estimation, where the first group is the "complete objects" with a size greater than 10 cm, such as orbiting satellites and abandoned complete payloads. The radius of these objects is uniformly assumed to be 1.9 m, and the mass is 1000 kg[23]. The second group is the space debris with a size of less than 10 cm. The radius of these debris is uniformly assumed to be 10 cm, and the mass is 1.2 kg[24].

(3) The calculation model of the average collision probability of LEO mega constellation in the shell adopts the "PIB" model[22]. The collision probability of the constellation in accordance with the density of space objects in the shell can be expressed as

$$CR_{O-D} = \pi(r_O + r_D)^2 V_{Rrl} \rho_O \rho_D V \tag{29}$$

where CR_{O-D} is the average collision probability of different targets, r_O is the outer envelope radius of the satellite, ρ_O is the satellite space density, V_{Rel} is the satellite relative target velocity, V is the shell volume, r_D is the space debris size, and ρ_D is the space debris density. The formula for calculating the density CR_{0-0} of the same space object in the shell is

$$CR_{0-0} = 2\pi r_0^2 \rho_0 V_{\mathrm{Re}l} (\rho_0 V - 1) \tag{30}$$

(4) Assuming that the mass and cross sectional area of the LEO mega constellation satellites are M_0 and A_0. The mass and cross sectional area of the new debris generated by the collision of constellation satellites in orbit satisfy the relational expression[25].

$$M_0[kg] = \begin{cases} 62.013 A_0^{1.13}, A_0 \geq 8.04 \times 10^{-5} m^2 \\ 2030.33 A_0^{1.5}, A_0 < 8.04 \times 10^{-5} m^2 \end{cases} \tag{31}$$

(5) The collision disintegration model meets the NASA collision disintegration model.

(6) The collision only produces two types of threatening space debris with a radius of 10 cm and a characteristic length sufficient to meet the collision disintegration energy and small debris that does not pose a threat to the disintegration of the satellite.

(7) Space debris collisions with a radius of less than 10 cm do not produce new satellite-threatening debris.

(8) The debris from the collision is evenly distributed within the shell.

(9) The average relative velocity of objects colliding in LEO is 10 km/s[15,18].

(10) Targets other than the constellation satellites in the shell are considered uncontrolled, and the controlled constellation satellites will not have a collision and disintegration event.

(11) The annual failure rate of constellation satellites due to events, such as microdebris collisions or equipment failures, is 2%.

3.2 Space object density calculation model

Spatial density refers to the statistical value of the number of space objects contained in each cubic kilometer of volume at a certain time and a certain spatial location, the unit is 1/km3, and the calculation method is as follows[26]:

$$s(x)=\lim_{\Delta V\to V^*}\frac{\Delta N}{\Delta V} \tag{32}$$

where V^* is the size of the volume element related to the distribution of space objects, and ΔN is the number of space objects in the space volume element. V^* needs to choose a reasonable value to reflect the distribution characteristics of space objects.

The above formula shows that that the spatial density is related to the size of the volume element and the number of objects in the volume element. However, the space objects move relative to the inertial space. At different times, the number of objects in the space volume element is different, the spatial density changes, and the low orbital target moves at a fast speed, so a certain instantaneous spatial density change has no practical reference importance. To better describe the density of space objects, *Öpik* and Wetherill et al[27]. introduced the concept of stay probability, averaged the instantaneous probability of space objects into the average probability over a period of time, and obtained the average density of space objects in the volume element as

$$s(x)=\lim_{\Delta V\to V^*}\frac{\Delta t}{T\cdot\Delta V} \tag{33}$$

where T is the orbital period of the space target, and Δt is the elapsed time of the space target entering and leaving the space volume element. On this basis, Kessler et al. made assumptions about the parameters of the target orbit and gave the calculation method of the spatial density of the object[28].

$$s(r)=\begin{cases}\dfrac{1}{4\pi^2 ra\sqrt{(r-r_p)(r_a-r)}};r_p<r<r_a\\ 0;r>r_a,r<r_p\end{cases} \tag{34}$$

$$f(\beta)=\frac{2}{\pi\sqrt{\sin^2 i-\sin^2\beta}},|\beta|<i \tag{35}$$

Where r is the geocentric distance; a is the semi major axis of the orbit; r_a and r_p are the geocentric distances of the orbit apogee and perigee, respectively. i is the inclination of the target orbit, and β is the declination of the target location. From the above formula, the spatial density of the space target at any height and latitude in space is obtained as

$$s(r,\beta)=s(r)f(\beta)=\frac{1}{2\pi^3 ra\sqrt{(\sin^2 i-\sin^2\beta)(r-r_p)(r_a-r)}} \tag{36}$$

This formula requires that $r_p<r<r_a$ and $\beta<i$. $s(r,\beta)=0$ when $r>r_a, r<r_p$ or $\beta>i$.

For the spatial density of the orbital height within the range of $[r_i^m, r_i^M]$, r_i^m and r_i^M represent the geocentric distance of the upper and lower boundaries of the ith height interval, respectively. Zhang derived the following four formulas for solving the spatial density by averaging over the height range[29].

1) When $r_p\neq r_a$ and $[r_i^m,r_i^M]\subseteq[r_p,r_a]$, the formula for calculating the spatial density is

$$s(r_i^m,r_i^M)=\frac{1}{4\pi^2 a[(r_i^M)^3-(r_i^m)^3]}\left\{\begin{array}{l}3a\left[\arcsin(2\dfrac{r_i^M-a}{r_a-r_p})-\arcsin(2\dfrac{r_i^m-a}{r_a-r_p})\right]-\\ 3\left[\sqrt{(r_i^M-r_p)(r_a-r_i^M)}-\sqrt{(r_i^m-r_p)(r_a-r_i^m)}\right]\end{array}\right\} \tag{37}$$

2) When $r_p\neq r_a$, $[r_i^m,r_i^M]\cap[r_p,r_a]\neq\varnothing$, and $r_i^M>r_a$, let $r_i^M=r_a$, and the formula for calculating the spatial density is

$$s(r_i^m,r_i^M)=\frac{3a\left[\dfrac{\pi}{2}-\arcsin(2\dfrac{r_i^m-a}{r_a-r_p})\right]+3\sqrt{(r_i^m-r_p)(r_a-r_i^m)}}{4\pi^2 a[(r_i^M)^3-(r_i^m)^3]} \tag{38}$$

When $r_p\neq r_a$, $[r_i^m,r_i^M]\cap[r_p,r_a]\neq\varnothing$, and $r_i^m<r_p$, let $r_i^m=r_p$, and the formula for calculating the spatial density is

$$s(r_i^m, r_i^M) = \frac{3a\left[\arcsin(2\frac{r_i^M - a}{r_a - r_p}) + \frac{\pi}{2}\right] - 3\sqrt{(r_i^M - r_p)(r_a - r_i^M)}}{4\pi^2 a[(r_i^M)^3 - (r_i^m)^3]} \tag{39}$$

3) When $[r_p, r_a] \in [r_i^m, r_i^M]$, let $r_i^M = r_a$ and $r_i^m = r_p$, and we obtain

$$s(r_i^m, r_i^M) = \frac{3}{4\pi^2 a[(r_i^M)^3 - (r_i^m)^3]} \tag{40}$$

If $e = 0$, then $a \in [r_i^m, r_i^M]$.

4) In other cases, $s(r_i^m, r_i^M) = 0$.

According to the density calculation formula, the density of space debris and satellites in the LEO region is calculated, and Figure 6 shows the curve of space target density with orbital altitude.

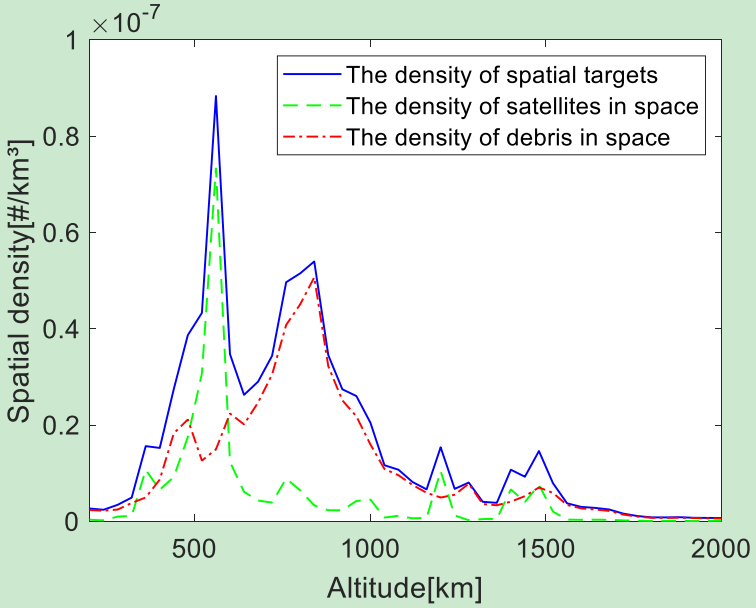


Figure 6. The density of LEO space targets varies with orbital altitude

3.3 Space debris attenuation model

Taking the shell as the research object, two sources of space debris are found, where one is the inflow from the upper shell, and the other is the collision of space objects. These space debris are continuously attenuated under the action of atmospheric resistance and gradually flow from this shell to the next. Therefore, the density of space debris in the shell changes dynamically with time. The schematic of space debris attenuation is shown in Figure 7.

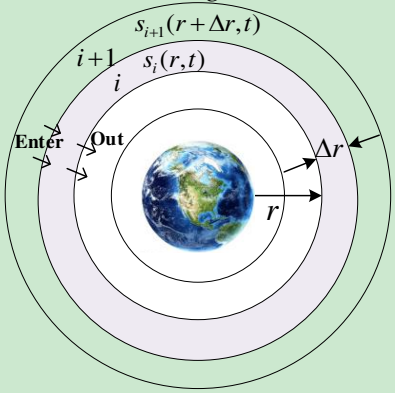


Figure 7. Schematic of space debris attenuation.

The decay rate of the same space debris in the shell is assumed to be the same as when the debris is in the center of the shell, and the atmosphere is stationary relative to the inertial frame. The orbital decay of space debris in a period can be integrated through the perturbation equation to obtain the average decay speed of the satellite in the shell. In accordance with the calculation formula of atmospheric drag acceleration, we have

$$\mathbf{a}_d = -\frac{1}{2} C_d \frac{A}{m} \rho |\vec{v}| \vec{v} \tag{41}$$

where C_d is the atmospheric drag coefficient, A is the target area, m is the target mass, ρ is the atmospheric density, and $\vec{v}$ is the velocity vector of space debris.

The atmospheric drag acceleration is projected along the velocity direction f_u, the geocenter radial direction f_n, and the orbital plane normal direction f_h. The projected components shown in Figure 8 are obtained.

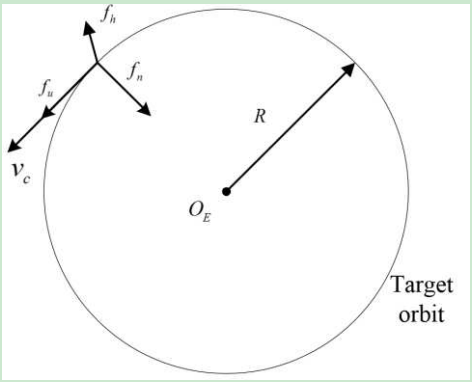


Figure8. Components of atmospheric drag along the direction of velocity, radial, and orbital surface normal.

Given that the velocity of the spacecraft along the radial vector direction and the normal direction of the orbital plane is zero, the components of atmospheric drag in the three directions can be obtained as

$$\begin{cases} f_u = -\frac{1}{2} C_d \frac{A}{m} \rho v^2 \\ f_n = 0 \\ f_h = 0 \end{cases} \tag{42}$$

This equation is brought into the Gaussian perturbation equation of motion

$$\begin{cases} \dot{a} = \frac{1}{n(1-e^2)^{1/2}} (1+e^2+2e\cos f)^{1/2} f_u \\ \dot{e} = \frac{2\sqrt{1-e^2}}{na} (1+e^2+2e\cos f)^{-1/2} (\cos f + e) \cdot f_u \\ \dot{i} = 0 \\ \dot{\Omega} = 0 \\ \dot{\omega} = \frac{2\sqrt{1-e^2}}{nae} (1+e^2+2e\cos f)^{-1/2} \sin f \cdot f_u \\ \dot{M} = n - \frac{\sqrt{1-e^2}(1+e^2+2e\cos f)^{-1/2}}{nae} (2\sin f + \frac{2e^2}{\sqrt{1-e^2}} \sin E) \cdot f_u \end{cases} \tag{43}$$

where E is the anomaly angle, $n = \sqrt{\mu / a^3}$ is the average angular velocity, and μ is the Earth's gravitational constant. The perturbation motion equation is averaged over the orbital period to obtain

$$\begin{cases} \bar{a} = \frac{1}{T} \int_0^T (\frac{1}{n(1-e^2)^{1/2}} (1+e^2+2e\cos f)^{1/2} f_u) dt \\ \bar{e} = \frac{1}{T} \int_0^T (\frac{2\sqrt{1-e^2}}{na} (1+e^2+2e\cos f)^{-1/2} (\cos f + e) \cdot f_u) dt \\ \bar{i} = 0, \bar{\Omega} = 0 \\ \bar{\omega} = \frac{1}{T} \int_0^T (\frac{2\sqrt{1-e^2}}{nae} (1+e^2+2e\cos f)^{-1/2} \sin f \cdot f_u) dt \\ \bar{M} = \frac{1}{T} \int_0^T (n - \frac{2\sin f}{nae} f_u) dt \end{cases} \tag{44}$$

In accordance with the Spacetrack website, 19,839 cataloged targets with an orbital height of less than 2000 km are found. The eccentricity of these cataloged targets can be counted, and the distribution of eccentricity along the semi major axis of the orbit can be obtained, as shown in Figure 9.

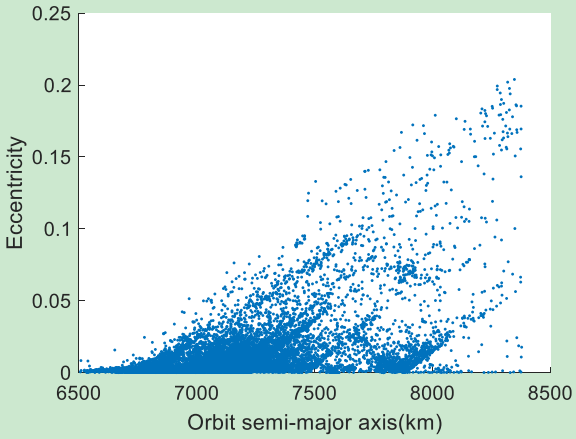


Figure 9. Distribution of eccentricity along the semi major axis of the track.

As shown in Figure 9, the eccentricity of most low-orbit targets is less than 0.05, and the space debris moves in approximately circular orbits in the low-orbit region. Therefore, the eccentricity of space debris can be set to zero, and the average perturbation motion of space debris in one cycle under the action of atmospheric resistance is

$$\begin{cases} \overline{a} = \dfrac{2}{n} f_u \\ \overline{M} = n \\ \overline{e} = 0, \overline{i} = 0, \overline{\Omega} = 0, \overline{\omega} = 0 \end{cases} \tag{45}$$

In accordance with equation (42), the average decay rate of space debris in the corresponding shell is

$$v_r = -\rho C_d \frac{A}{m} \sqrt{\mu a} \tag{46}$$

The static atmospheric resistance ρ in the above formula is determined by the exponential model

$$\rho = \rho_0 e^{-\frac{r - r_0}{H_0 + \frac{\chi}{2}(r - r_0)}} \tag{47}$$

where $\chi \approx 0.1$, $H_0 = 37.4km$, $r_0 = H_0 + 6378.137$, and $\rho_0 = 3.6 \times 10^{-10} kg \cdot m^{-3}$.

The decay rate of space debris is relatively slow, and the height of the far and near points does not change much during the 7-year simulation time due to the large difference in the height of the far and near points of the large elliptical orbit, so the influence of the space density in this part of the space debris shell is ignored.

3.4 Space target collision disintegration model

During the long-term evolution of the constellation shell, space objects in the shell may collide, and collisions between different space objects will produce different amounts of space debris, causing different effects on the environment. Space debris is classified in accordance with different sizes to study the specific influence of the collision of different space objects on the space environment, and the amount of space debris generated by collisions between different types of space objects and the distribution of debris in space are analyzed.

(1) Collision type investigation

In this study, the space debris is divided into two types: cataloged debris and complete objects. In accordance with the classification, six types of collisions between space objects, which are collisions between cataloged debris and collisions between cataloged debris and space intact objects, can be obtained to catalog collisions between fragments and loads, collisions between space intact objects, collisions between space intact objects and loads, and collisions between loads and loads. In accordance with the close encounter events of space objects published on the Spacetrack website, 1086 events were selected for statistical analysis, and the distribution of different types of collision events was studied[19]. Figure 10 shows the statistics of the rendezvous target type and the number of rendezvous.

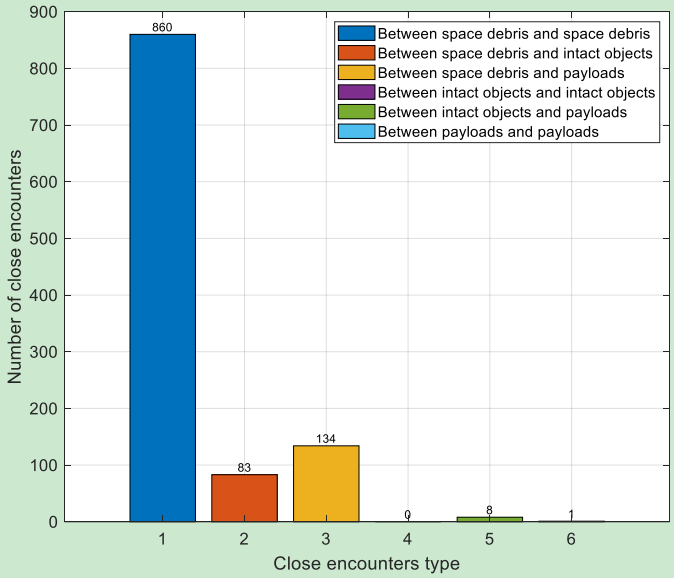


Figure 10. Rendezvous target type and rendezvous statistics.

In accordance with the statistics in Figure 10, the close encounters between cataloged fragments account for 79.19% of the total number of encounters, the collision between cataloged fragments and complete objects is 7.64%, and the collision between cataloged fragments and loads is 12.34%. The collision between loads is 0.74%, and the collision between loads is 0.09%. Therefore, the proportion of the collision probability between various types of space objects to the total collision probability is assumed as follows: the collision probability between cataloged fragments accounts for 79.19% of the total, the collision probability between cataloged fragments and complete objects accounts for 19.98% of the total proportion, and the collision probability between complete objects and complete objects accounts for 0.74% of the total proportion.

(2) Spacecraft breakup model

The amount of debris produced by the collision of different types of space targets is different. This condition is mainly because the collision energy is different. The calculation of the collision energy can be expressed as

$$E_P = \frac{m_p v_r^2}{2m_t} \tag{48}$$

where E_p represents the collision energy, m_p is the target with smaller mass, m_t is the target with larger mass, and v_r is the relative motion speed. Under the condition of the same relative collision speed, the collision energy is mainly related to the mass between the two collision targets. When the collision energy is small, the collision will not cause the target to disintegrate. At present, the two targets can be completely disintegrated when the energy generated by the collision of the center of mass is greater than $40kJ/kg$. The collision and disintegration model of space targets adopts the American NASA collision disintegration model. In accordance with the equivalent mass of the two colliding targets, the number of space debris generated with different characteristic lengths can be estimated as

$$N_F(l \geq l_c) = 0.1\hat{m}^{0.75} l_c^{-1.71} \tag{49}$$

where N_{CF} is the number of space debris larger than the characteristic length l_c, the characteristic length is $l_c = (l_x + l_y + l_z)/3$, l_x, l_y, l_z are the projected sizes of the space debris on the three orthogonal principal axes, and d is the equivalent mass of the two targets, which is expressed as

$$\hat{m} = \begin{cases} \dfrac{m_t + m_p}{[kg]}, E_p \geq E_p^* \\ \dfrac{m_p v_r^2}{[kg][km/s]^2}, E_p < E_p^* \end{cases} \tag{50}$$

4 Simulation results and analysis

Orbital collisions are divided into short-term natural evolution orbital collisions and long-term orbital collisions during the constellation lifetime to evaluate the influence of constellations in different states, the TLEs obtained from Reference[19]. This study divides the space into multiple orbital layers to facilitate the calculation. Given that the mega constellation satellites in LEO operate at the near designed orbit, only the collision in the orbital layer where the mega constellation is located needs to be calculated.

4.1 Simulation and analysis of short-term collision probability

4.1.1 Numerical experimental setup

The short-term collision probability orbit prediction model uses the SGP4 model to simulate the first phase of the Starlink constellation and to facilitate the prediction of the orbit of the space cataloging target[30]. The constellation parameters of the first phase of Starlink are Walker-δ 1584/72/33, the orbit inclination is 53°, and the orbit height is 550 km. The shell height is 40 km, the constellation shell interval is the orbit height 530—570 km area, and the simulation time is 7 days.

All cataloged targets on March 12, 2022 are downloaded through the Spacetrack website[19], and the apogee screening method is used to preliminarily screen the space targets. The targets with perigee greater than 570 km and the selected position less than 530 km are excluded. A total of 4829 space targets will be combined with the intersection of the height intervals of the constellation shells. Taking the space debris as the statistical interval of 100 km, the orbital semi major axis distribution of 4,829 space targets is counted. Figure 11 shows the distribution of the number of space targets in accordance with the orbit semi major axis after screening.

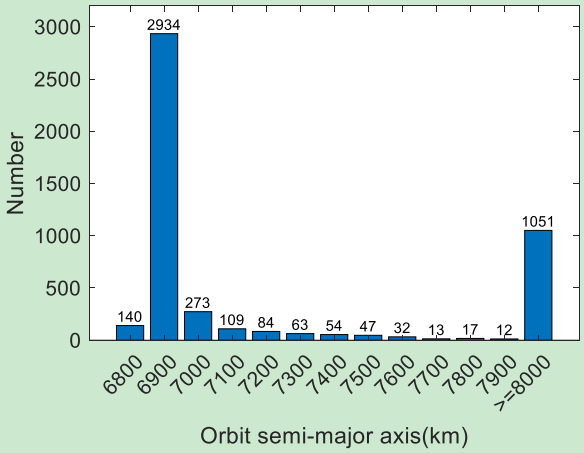


Figure 11. Distribution map of the number of space targets remaining in the apogee–perigee screening along the semi major axis of the orbit.

The remaining space targets are processed, and the TLE data of all satellites and space debris are converted by using STK after the invalid data are eliminated by the apogee screening method. The space targets within the shell are examined, and 653 space targets are likely to be close to space debris for a long time. The SGP4 model is used to predict the orbits of these targets and the low-orbit giant constellation satellites and long-term space targets in the shell, and the simulation step size is set to 1 s. The simulation step size is set to 5 s for the remaining 4176 space targets, and the time interval closest to the shell is solved in accordance with the orbit height, and the simulation step size of this interval is set to 1 s to perform the orbit evolution.

A total of 4176 space target orbits are screened in terms of height by using the evolved orbit data. The arcs with the height of the space target orbits in the interval of 530–570 km and their latitudes are examined. Given that the orbital inclination of the Starlink satellite is offset by about 0.054° and the random deviation between the collision target and the collision position of the satellite is considered, the latitude screening value is set to 54°.

The space discrete volume element is used to perform a preliminary approach analysis after obtaining the arcs where the space target may collide with the low-orbit giant constellation. The included angle between the adjacent orbital planes of the Starlink constellation is 5°, and the included angle between adjacent satellites in the orbital plane is about 16.3°. Therefore, the trajectory of the space target in the space volume element should be increased as much as possible to minimize the number of satellite targets in the space. The values of space volume elements $\Delta h, \Delta\lambda, \Delta\varphi$ are set to 40 km, 4.8°, and 9°, respectively. In accordance with the space targets that appear in the same space volume element at the same time, the relative distances are calculated pairwise by using the orbit data, and the distance to the closest trajectory point and the "3σ" distance are judged to remove possible collision targets.

4.1.2 Results and analysis

In accordance with the filtered data, a large part of the space targets is close to each other for a long time. For this part of the target data, the orbital period of the space target is used to calculate the closest distance of the trajectory point in each period, and the closest distance to the trajectory point and its vicinity are considered. This trajectory point is the most likely to collide, and other trajectory points will not collide, thereby eliminating other trajectory points in each orbital cycle of the target. The approaching situation of the space target within 7 days is obtained after eliminating the long-term approaching target data. Figure 12 shows the statistics for different types of proximity situations.

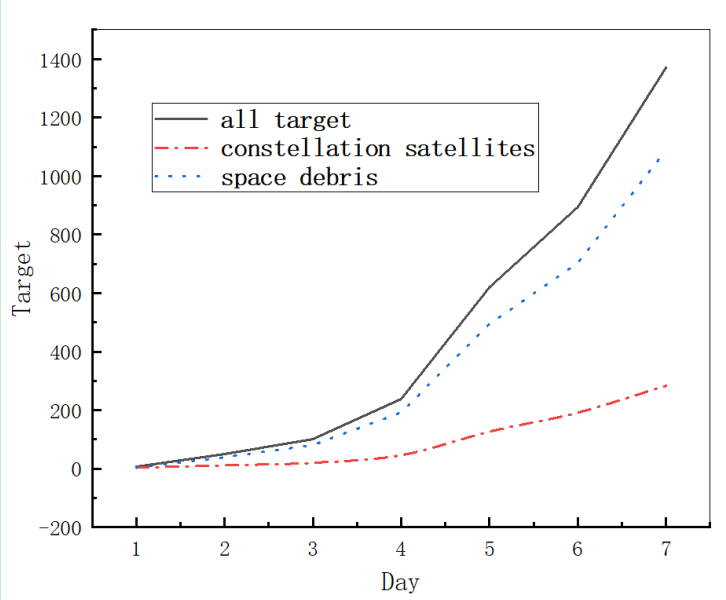


Figure 12. The statistics for different types of proximity situations

Table 2. Close proximity of space targets in the shell.

Close target type	Day one	Day two	Day three	Day four	Day five	Day six	Day seven
Number of times close to all target	7	50	101	238	620	894	1371
Number of times close to constellation satellites	3	11	20	46	126	191	283
Number of closes between space debris	4	39	81	192	494	703	1088

As shown in Table 2, many close points will be missed when the ellipsoid error is small due to the small relative screening distance. With the gradual increase in error, the relative screening distance and the number of targets that meet the proximity conditions increase. The time of the

closest track point retained is filtered in accordance with the relative distance, and two points are selected before and after the track point. Lagrangian interpolation is used to calculate the closest moment and the motion state of the two targets at that moment and bring them into the Laplacian-based method. The collision probability calculation method of the Si transform is used to obtain the collision risk situation of the space target. Figure 13 shows collision probability of space rendezvous target greater than 10^{-5} times

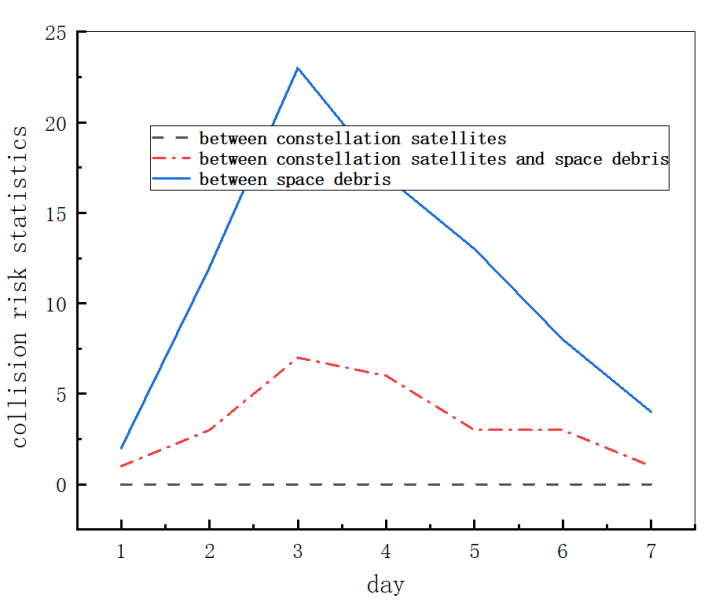


Figure 13. Collision probability of space rendezvous target greater than 10^{-5} times

Table 3. Collision probability of space rendezvous target greater than 10^{-5} times.

Collision target type	Day one	Day two	Day three	Day four	Day five	Day six	Day seven
Between constellation satellites	0	0	0	0	0	0	0
Between constellation satellites and space debris	1	3	7	6	3	3	1
Between space debris	2	12	23	17	13	8	4

As shown in Table 3, the number of collisions between constellation satellites and space debris greater than 10^{-5} reaches 7 times within the 7-day simulation period The collision risk increases by about 30% to 40% after the constellation is deployed compared with before the deployment of the constellation by comparing the number of constellation satellites with a collision probability greater than 10^{-5} and the number of collisions between space debris with a probability greater than 10^{-5}.

4.2 Simulation and analysis of the long-term collision probability

4.2.1 Numerical experimental setup

The long-term collision probability simulation calculation of the first phase of the Starlink constellation is performed. The known constellation parameters are Walke-δ1584/72/33, the orbit height is 550 km, the orbit inclination is 53°, the satellite mass is 260 kg, and the satellite size is $3.2m\times1.6m$. Let the simulation time be 7 years and the constellation shell height H=40 km.

Considering that the atmospheric resistance can be approximately ignored when the orbit height is greater than 800 km, the shell is divided into 7 intervals in accordance with the orbit height of the Starlink constellation, and the interval height range is 530—810 km. From Assumption 6 and Formula (48), the mass of space debris c that reaches the collision disintegration energy is 0.208 kg. From Assumption 4, the cross sectional area of the space debris of type c can be obtained as 0.0065 m2, and

the radius of the space debris of type C can be obtained as 0.045 m. In accordance with the definition of the characteristic size, the attenuation speed of the space target in each shell can be obtained by using the static atmospheric drag density model after the surface-to-mass ratio and orbit height of the space target are known. Figure 14 shows the different decay rates of the target in each shell.

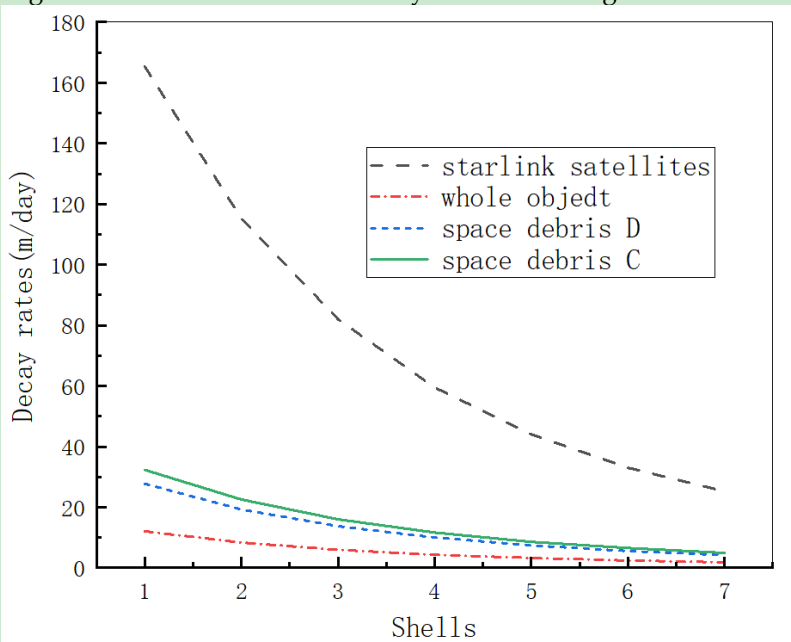


Figure 14. The different decay rates of the target in each shell

Table 4. Relationship between the decay velocity and the shell of the different types of space targets .

Target type	Shell one (m/day)	Shell two (m/day)	Shell three (m/day)	Shell four (m/day)	Shell five (m/day)	Shell six (m/day)	Shell seven (m/day)
Starlink satellites	165.4	115.2	82	59.5	44	33	25.2
Whole object I	12	8.3	5.9	4.3	3.2	2.4	1.8
Space debris D	27.7	19.2	13.7	10	7.4	5.5	4.2
Space debris C	32.3	22.5	16	11.6	8.58	6.5	4.9

The space debris environment in the shell can be obtained from the existing cataloging data. The cataloging target data can be downloaded from the Spacetrack website, and the download time is March 12, 2022. The downloaded cataloging targets are screened in terms of the shell, and each item at the initial moment can be obtained. The number of space debris is cataloged in the shell. Figure 15 shows the distribution of the number of space debris in each shell.

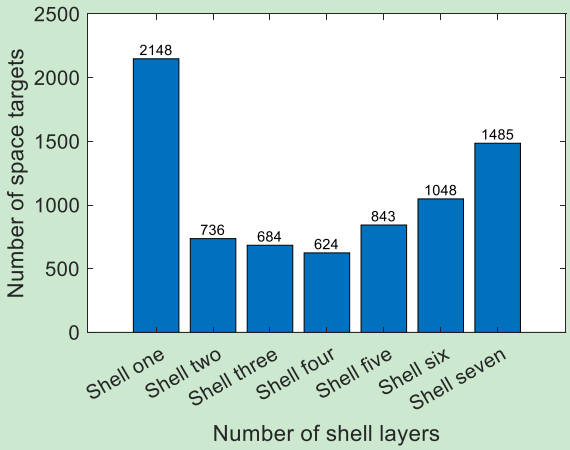


Figure 15. Distribution map of the number of space debris in each shell.

In accordance with the statistics of literature [8], the number of rocket bodies in the low-orbit area is 916, and the number of loads is 6420 as of April 4, 2022. Assuming that the distribution of rocket bodies is proportional to the number of loads, Starlink can be obtained through the Spacetrack website. The number of satellites in the first shell is 1495, and the number of other payloads and space debris in the first shell is 653. Therefore, the number distribution of the rocket body in each shell and the statistics on the number of satellites in each shell can be obtained, and the number of complete objects in each shell can be calculated. Table 5 shows the distribution of complete debris types *I* and *D* in the shell.

Table 5. Quantity distribution of the different types of space debris in the crust.

Target type	Shell one	Shell two	Shell three	Shell four	Shell five	Shell six	Shell seven
All fragments	653	736	684	624	843	1048	1485
Whole object *I*	371	423	193	117	135	130	306
Space debris *D*	282	313	491	507	708	918	1179

4.2.2 Results and analysis

The number of space debris in the shell is the number of debris in this shell plus the number of debris entering the higher shell and the number of debris falling out of this shell. In accordance with Tables 4 and 5, the amount of space debris in the constellation life cycle of the first shell due to orbital decay changes as follows:

$$\begin{cases} N = 653 - \dfrac{N_{1,I}}{40000} v_{1,I} t - \dfrac{N_{1,D}}{40000} v_{1,D} t + \dfrac{N_{2,I}}{40000} v_{2,I} t + \dfrac{N_{2,D}}{40000} v_{2,D} t, t \le 1444 \\ N = N_{1444} + (\dfrac{N_{2,I}}{40000} v_{2,I} + \dfrac{N_{2,D}}{40000} v_{2,D} - \dfrac{N_{1,I}}{40000} v_{1,I} - \dfrac{N_{2,D}}{40000} v_{2,D})(t - 1444), t \le 2083 \\ N = N_{2083} + (\dfrac{N_{2,I}}{40000} v_{2,I} + \dfrac{N_{3,D}}{40000} v_{3,D} - \dfrac{N_{1,I}}{40000} v_{1,I} - \dfrac{N_{2,D}}{40000} v_{2,D})(t - 2083), t \le 2919 \end{cases} \quad (51)$$

where N is the number of debris, subscripts i, j represent the j type of the i-th shell fragment, v is the decay speed, and t is the number of days. In accordance with the definition of density $\rho = N / V$, the evolution relationship of the space environment density with time during the life cycle of the Starlink constellation can be obtained. Figure 16 shows the spatial density evolution of background debris in the crust.

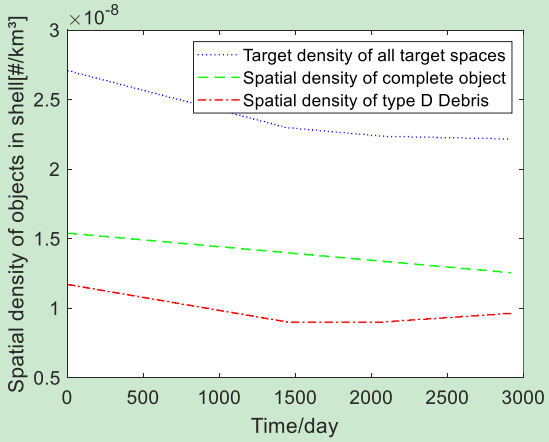


Figure 16. Spatial density evolution of background debris in the crust.

In accordance with the assumption, the space debris only collides with the failed Starlink satellites. Up to 32 satellites fail each year at the failure ratio of 2%. Given that the satellites decay at a rate of 165.4 m per day in the shell, the satellites are assumed to be evenly distributed in the shell. In 4 months, half of the satellites fell out of the shell due to attenuation. Although 32 satellite failures occurred in the first and second year, the spatial density of the failed satellites reached a maximum of 2.6526×10^{-9}, which was still lower than the background. If no collision event occurred in space during the lifetime of the constellation, the spatial density is reduced by 4%–16.7% compared with that before the deployment of the constellation. Starlink satellites and intact objects generate new space debris due to collision. Therefore, the collision disintegration events are divided into six types, and the amount of space debris generated by different collision events is calculated. Figure 17 shows the amount of space debris generated by different collision types.

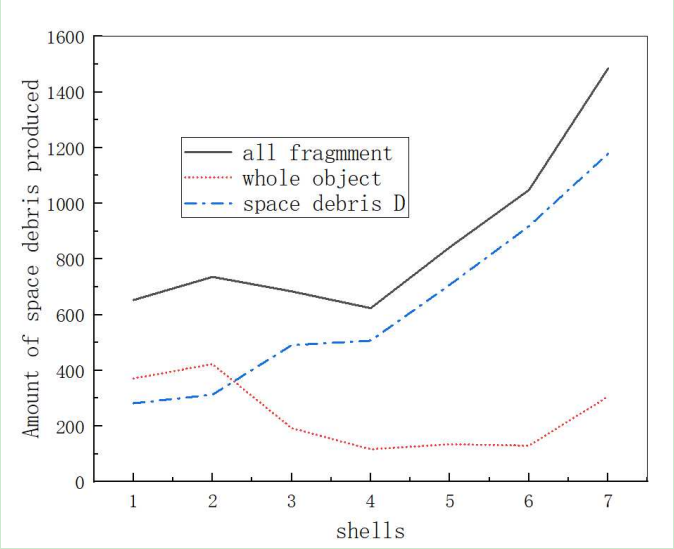


Figure 17. The amount of space debris generated by different collision types.

Table 6. Amount of space debris produced by different collision types.

Collision target type	Space debris D number	Space debris C
Starlink satellites and Intact Objects	624	17
Starlink satellites and space debris D	65	132
Starlink satellites and space debris C	64	132
Whole objects and whole objects	1643	385
Intact objects and space debris D	466	74
Intact objects and space debris C	465	74

As shown in Table 6, a large amount of debris are generated in one collision, the least one collision generates approximately 200 space debris, and the density increases. In accordance with the average

value of the debris generated, the generated space debris reaches 658, the density increases, and the space debris increases. The density is doubled when the constellation is not deployed. In accordance with Table 4, the generated space debris has a small surface mass ratio, a low orbital decay rate, and the debris has a long residence time in the orbit, resulting in a long-term collision of the constellation in orbit. The risks are multiplied, and in-orbit collisions with satellites should be avoided as much as possible.

Taking the Starlink constellation as an example, the short-term collision probability and long-term collision probability of the constellation are calculated, and the results are as follows:

(1) The probability of collision between Starlink Phase I constellation satellites and space debris is greater than seven times a day. Compared with before the deployment of the constellation, the collision probability in the shell increases by about 30%–40%, indicating that the collision risk faced by the satellite is higher than that before the deployment of the constellation. The satellite needs to have better orbital maneuvering and collision warning abilities.

(2) During the 7-year evolution period of the Starlink constellation, the average collision probability after the deployment of the Starlink constellation does not increase compared with that of the nondeployed constellation, indicating that a lower constellation orbit height can improve the orbital safety and space environment stability of the low-orbit giant constellation.

(3) The occurrence of a collision greatly increases the density of space debris, and the space debris generated by the collision is small and stays in orbit for a long time, resulting in a greater threat of secondary collisions to constellation satellites.

5 Conclusion

This paper investigates the collision safety of LEO mega constellations under the current space debris environment. A collision evolution calculation method between large-scale space debris and large-scale satellites is proposed. By filtering and partitioning data, the computing efficiency is improved while satisfying high accuracy.

The lower orbital altitude of the Starlink constellation improves the safety of the constellation and the stability of the space environment, but the collision risk faced by the constellation is still high. The satellite needs to have good orbital maneuverability and collision warning capabilities. The risk of a second collision in a constellation increases exponentially.

This study has proposed a calculation method for collision safety assessment parameters of LEO mega constellation. The short-term collision probability and long-term average collision probability of the constellation are used to evaluate the safety of the constellation to improve the reliability and effectiveness of the constellation safety assessment. Through the simulation of Starlink phase I constellation, results show that after the deployment of the Starlink constellation, the probability of a short-term collision in the constellation shell increases by 30%–40%, the probability of at least one collision in the constellation lifetime is 70.2%, and the probability of a secondary collision increases by 25.3% after a collision.

The main topics for possible future improvements. Space debris had become a growing problem in the space sector, and the deployment of large constellations consisting of hundreds or thousands of satellites was an environmental concern. The occurrence of a collision causes more serious consequences. The method of how to avoid collision of constellation satellites and improve the safety of constellation in-orbit operation should be further investigated.

Data Availability Statement: The data are available from the corresponding author on reasonable request. Data access is restricted to research and educational applications.

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